# DocAI Hardware & Material Guide: Substrate Analysis and Luminescence

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## 1. Optical Brightener Free Security Paper

Genuine security paper for passports contains no optical brightening agents (OBAs). Under ultraviolet illumination (365 nm UV-A), authentic paper remains dull dark, whereas commercial copy paper fluoresces bright blue-white due to optical whitening chemicals.

## 2. Polycarbonate Thermal Fusion

Polycarbonate identity cards consist of multiple thin foil layers laminated under heat and pressure into a solid monolithic block. Once fused, individual layers cannot be separated without irreversible destruction, rendering mechanical knife-splitting counterfeiting impossible.